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**Q1** Describe the Joux's three-party Diffie-Hellman protocol

**Q2** What is the (min) order of the following multiplicative groups defined by the chosen parameters  $p$  and  $g$ :  
[hint: 593, 1187, 3559 primes, with  $1187 = 2 \times 593 + 1$  and  $3559 = 593 \times 2 \times 3 + 1 \rightarrow$  with the exception of (2), no need to do any modular exponentiation! – **NOTE: EXPLAIN YOUR ANSWER otherwise answer is not valid**]

1)  $g^x \bmod p$ , with  $g=1186$ ,  $p=1187$

Order 2: 1186 is -1

2)  $g^x \bmod p$ , with  $g=7$ ,  $p=1187$

Must check  $7^{593} \bmod 1187 \rightarrow -1 \rightarrow g$  is generator of the full group, order 1186

3)  $g^x \bmod p$ , with  $g=9$ ,  $p=1187$

$9 = 3^2 = 3^2 \bmod 1187$ ,  $g$  is QR, order is  $(p-1)/2 = 593$

4)  $g^x \bmod p$ , with  $g=64$ ,  $p=3559$

Just note that  $64 = 2^6 \rightarrow$  order is  $(p-1)/6 = 593$

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**Q3** Describe the Pedersen Verifiable Secret sharing

**Q4** Being  $e: G \times G \rightarrow G$  a bilinear map, and  $g$  a generator of  $G$ , simplify the expression:

$$e(g^a g^b, g^c g^d) / e(g^{ac}, g^{bd})$$

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**E1.** Consider the Elliptic curve  $y^2 = x^3 - 2x - 3$  defined over the modular integer field  $\mathbb{Z}_5$ .

**A.** find all the points  $EC(\mathbb{Z}_5)$  and state what is the order of the corresponding group

$$\begin{aligned} P &= (x_1, y_1) \\ Q &= (x_2, y_2) \\ R &= P + Q = (x_3, y_3) \\ x_3 &= \lambda^2 - x_1 - x_2 \\ y_3 &= \lambda(x_1 - x_3) - y_1 \\ \lambda &= \begin{cases} \frac{y_2 - y_1}{x_2 - x_1} & P \neq Q \\ \frac{3x_1^2 + a}{2y_1} & P = Q \end{cases} \end{aligned}$$

**B.** Compute  $[4](1,1)$

*[HELP: possibly useful mnemonic hints reported here on the right;*

*MUST-DO: show step-by-step detailed computations; try to minimize the number of EC additions]*

**points: 0, (1,1), (1,4), (2,1), (2,4)**

**order: 5**

**$[4](1,1) = [2]((1,1)+(1,1)) = [2](2,1) = (2,1)+(2,1) = (1,4)$**

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**E2 – Secret Sharing and Threshold RSA.** A group of four parties forms a (3,4) secret sharing scheme. Using ordinary arithmetic (no moduli) parties  $P_i = \{1,2,3,4\}$  are dealt with shares  $\sigma_i$  of a key SK, which is the private key of an RSA scheme with modulus  $n=91$  and public key  $PK=5$ .

**A.** Using shares  $\sigma_1=34$ ,  $\sigma_2=41$ ,  $\sigma_4=61$ , reconstruct the secret key SK and verify that it is the correct one for the considered RSA public key PK;

$\text{Lambda1}=8/3$ ,  $\text{lambda2}=-2$ ,  $\text{lambda4}=1/3$ ;

$8/3*34-2*41+1/3*61 = 29 \rightarrow \text{OK because } 29 = \text{PowerMod}[5,-1,72] \text{ being } 72 = \text{EulerPhi}[91]$

**B.** If the attacker only knows shares  $\sigma_1$  and  $\sigma_4$ , and does not yet know share  $\sigma_2$ , what's his/her advantage in terms of chances to guess the secret key versus a pure random guess?

Double chance to guess, as the secret key must be odd: being  $x$  the unknown value of the share,  $8/3*34-2*x+1/3*61=111-2x$  hence  $sk$  must be odd.[note: we are NOT accounting for the fact that  $sk$  is an RSA key so it also must be odd... i.e. strictly speaking in this very specific case there is no real advantage]

**C.** Using the Shoup construction, show step by step how the three parties  $P_1$ ,  $P_2$  and  $P_4$  can distributely compute a threshold RSA signature for the message  $m=15$ , and verify that the result is correct by directly computing the signature using the SK value computed before.

[NOTE: if you prefer, to simplify computation instead of using the full  $L!$  you can use the minimum value that permits to make all lambda coefficients integer] [HINT:  $15^{-1} \bmod 91 = 85$ ]

Since lambda denominators are at most 3, each party computes  $m^{(3*\text{lambda}_i*\sigma_i)} \bmod 91 = \{22,64,15\}$ .  
Multiplying mod 91,  $m^{(3d)} = 8$ . But we also know that  $m^{ed} = m^{(59d)} = m = 15$ .

Now we can apply the common modulus attack:  $\text{ExtendedGCD}[59,3] = \{-1,20\}$ . Hence

$M_{\text{signed}} = 8^{20} * 15^{(-1)} \bmod 91 = 64 * 85 \bmod 91 = 71 \rightarrow \text{check: } \text{PowerMod}[15,29,91]=71 \text{ OK}$